PRISM: Personalized Prediction of Long Screen-Off Intervals on Smartphones

Yichen Peng
High School Student
The High School Affiliated to Renmin University of China
yichen_peng@outlook.com

Shuaiming Zhu, Changhua Pei, Jingjing Li
Computer Network Information Center
Chinese Academy of Sciences
zhushuaiming26@mails.ucas.ac.cn
chpei@cnic.cn, ljj@cnic.cn

Xiao Liang*
The High School Affiliated to Renmin University of China
3014640435@qq.com

***Abstract*—Accurate prediction of long screen-off intervals could enable deep standby on smartphones. We study 4.9 million screen-off events from a leading smartphone company and predict whether the corresponding screen-off interval, ending at the next screen-on event, exceeds four hours. Prediction is challenged by sparse context, individual differences, and behavioral drift. We evaluate two protocols. Immediate prediction invokes the model at screen-off, whereas delayed prediction invokes it only if the screen remains off after a scenario-dependent invocation delay. We propose PRISM, a model for personalized prediction of long screen-off intervals on smartphones. PRISM combines jointly learned user-profile and user-history embeddings with scenario, screen-off reason, and cyclic-time embeddings, explicitly modeling user-profile–scenario and scenario–reason interactions. A user-history residual subnetwork uses the updated, uncompressed history vector to refine the prediction logit. With approximately 35k trainable parameters, PRISM performs single-event inference about $1.8\times$ and $3.3\times$ faster than the LightGBM and Transformer baselines, respectively. It also achieves the highest accuracy and F1 scores in both the immediate and delayed prediction settings on the chronological test set.**

I. Introduction

A *long screen-off interval* is a continuous period from a smartphone's screen-off event to its next screen-on event that exceeds a predefined threshold, set to four hours in this study. We predict whether an interval will exceed this threshold using information available at decision time. Accurate prediction could enable deep standby and reduce background energy consumption, while avoiding premature standby and recovery overhead when the user returns sooner.

Predicting such intervals is challenging. Related work explores short-term unlocking prediction with manually selected contextual features and conventional classifiers [1], but reliable multi-hour screen-off prediction requires addressing three difficulties. First, the available context is sparse. The screen-off reason distinguishes manual actions, inactivity timeouts, and other causes, while the timestamp identifies a scenario such as daytime or night. Second, behavior varies substantially across individuals and remains uncertain within a given scenario. Third, an individual's behavior may drift. Changes in employment, routines, or seasons may introduce long-off periods that were previously rare. Static profiles and fixed population-wide rules cannot directly reflect these changes.

*Corresponding author.

We propose PRISM – a Personalized Reason-Individual-Scenario Model for long screen-off prediction. Its three complementary design components address these challenges.

First, dedicated user-profile and user-history subnetworks learn representations of completed events jointly with the prediction network. These representations capture overall tendencies, recent activity, and context-specific behavior. PRISM retains engineered history summaries but learns their representations and combinations end to end, reducing reliance on manually designed feature combinations.

Second, PRISM explicitly models user-profile–scenario and scenario–reason interactions through element-wise products of learned embeddings. A scenario's predictive meaning can thus depend on the user, while a reason's meaning can depend on the scenario. These lightweight interactions encode contextual dependencies without large crossed lookup tables.

Third, a user-history residual subnetwork processes the uncompressed history vector and adds a correction to the main prediction logit. This direct path preserves signals that may be weakened by the compact history embedding. Updating the history after completed events lets recent behavioral changes influence subsequent predictions, providing a mechanism for responding to changing routines.

We evaluate PRISM on 4.9 million real-world events from 53,122 users with a chronological test split. Immediate and delayed predictions are made at screen-off and after a scenario-dependent delay. PRISM achieves the highest accuracy and F1 in both settings: accuracy reaches 88.71% and 92.89% while F1 reaches 82.96% and 89.50%. With 35,346 parameters, inference takes approximately 0.4 ms per event, about $1.8\times$ and $3.3\times$ faster than LightGBM and Transformer, respectively. Ablations support the design choices, and contextual analysis locates the clearest gains in Weekday Day and Weekend.

II. Related Work

Mobile behavior prediction. Screen-off prediction connects user-behavior modeling with mobile energy management. Unlike offline duration classification, deployment must decide both whether an interval will be long and when to predict it. We therefore evaluate immediate and delayed prediction using only information available at decision time.

Tabular and sequence models. Categorical Bayes provides an interpretable low-capacity reference, while LightGBM [2] represents a strong tree-based model for fixed history summaries. Recurrent networks [3] and Transformers [4] model ordered event histories. We include both LightGBM and Transformer to compare explicit history statistics with sequence encoding.

Feature interactions. Factorization machines [5] model pairwise feature interactions without enumerating every crossed category. PRISM adopts this principle for screen-off prediction: user profile, scenario, and reason are represented separately, and only the empirically supported embedding products are retained.

III. Task and Protocol

Dataset and protocol. The dataset spans 2025-10-01 to 2026-02-10 and contains 4,894,616 screen-off events from 53,122 users; 31.96% exceed four hours. Each event contains an anonymized user id, screen-off and next screen-on timestamps, current screen-off reason, and next screen-on reason. Event i's next-screen-on reason is future information for event i but becomes visible to later user-history features after that interval completes. We use a timestamp-sorted 70/10/20 split: the first 70% fits models, the next 10% validates epochs, hyperparameters, thresholds, and variants, and the final 20% forms the test set. All event-level features are computed using only information available at the time of prediction.

Let d_i denote the screen-off duration of event i, s_i its scenario, and $\tau(s_i)$ the corresponding invocation delay. Under the delayed prediction policy,

$$\hat{y}_i = \begin{cases} 0, & d_i < \tau(s_i), \\ \mathbb{I}[\hat{p}_i \geq \theta], & d_i \geq \tau(s_i), \end{cases} \tag{1}$$

where $\hat{p}_i$ is the predicted positive probability and θ is a global threshold selected on validation data. Immediate prediction sets $\tau(s) = 0$, so every event is scored immediately. The four-hour target refers to the total interval duration measured from screen-off, rather than from model invocation.

Metric. The logs lack calibrated energy savings, transition cost, recovery cost, and device-state overhead, so we use positive-class F1 as the primary metric: precision tracks unnecessary positive decisions and recall tracks missed long-off opportunities. Accuracy, precision, and recall are reported alongside F1; F1 is not treated as measured energy gain.

IV. PRISM Architecture

This section presents the technical design of PRISM, illustrated in Figure 2.

Model overview. PRISM combines the history-derived E_u and E_h with scenario, reason, and cyclic-time embeddings E_s, E_r, and E_t. The prediction subnetwork uses their selected interactions and a direct history residual before the sigmoid and validation-selected global threshold. The components are detailed below.

Scenario design. The scenario partition enables delayed prediction to use different invocation delays across behaviorally distinct time regimes. We consider four interpretable scenarios aligned with daily routines: Night, Weekday Day (working hours), Weekday Evening (after-work hours), and Weekend.

We determine their exact boundaries in two constrained steps. Let $\bar{y}(W)$ denote the empirical positive rate within window W. A 30-minute sweep over circular 8-hour windows first identifies the shared Night period:

$$N^* = \arg\max_{W \in \mathcal{W}_8} \bar{y}(W) = [23{:}00, 07{:}00). \tag{2}$$

With Night fixed, we sweep candidate boundaries a between Weekday Day and Weekday Evening at 30-minute intervals. For each candidate, we estimate the Bernoulli positive rate of each scenario and select the boundary with the highest total log-likelihood:

$$\begin{aligned} \ell(a) &= \sum_g \left[n_g^+ \log \hat{p}_g + n_g^- \log(1 - \hat{p}_g) \right], \\ a^* &= \arg\max_{a \in \mathcal{A}} \ell(a) = 17{:}00, \end{aligned} \tag{3}$$

where n_g^+ and n_g^- are the positive and negative counts in scenario g, and $\hat{p}_g = n_g^+/(n_g^+ + n_g^-)$. A deterministic router assigns each screen-off event to one of the four scenarios using its time of day and day of week. Each scenario's 16-D embedding is learned jointly with the model. We use a 0.5 h invocation delay for Night and Weekday Evening and 2.5 h for Weekday Day and Weekend, invoking PRISM earlier in more predictable regimes while allowing short intervals in less predictable regimes to end before invocation.

User modeling. PRISM forms six user-history features covering activity volume, long-off tendency, temporal preference, and recency. These features include log event count, historical positive rate, and day/night activity-density share, together with three statistics describing recent behavior. After LayerNorm, a GELU-activated $6 \to 16 \to 16$ subnetwork maps them to the user-profile embedding E_u. The six features are updated only after each event completes.

Reason and cyclic-time embeddings. The observed screen-off reason indexes a trainable embedding table to obtain E_r. To retain fine-grained time information within each scenario, a cyclic-time subnetwork maps $[\sin\phi_m, \cos\phi_m, \sin\phi_w, \cos\phi_w]$ to E_t, where $\phi_m = 2\pi m/1440$ and $\phi_w = 2\pi w/7$ denote minute-of-day and day-of-week phases. This encoding preserves continuity across midnight and week boundaries.

Interaction design. The base embeddings represent individual factors but do not encode their interactions. PRISM therefore introduces Hadamard products and uses the empirical positive-rate surfaces in Figure 1 to select the relevant pairs.

The figure considers the four scenarios, the five most frequent reasons, and four visualization-only user-profile groups. These groups are obtained by binarizing day/night activity density and historical positive rate and crossing the two binary labels; they serve only as discrete proxies for the continuous user-profile embedding E_u.

Figure 1 shows that the user-profile–scenario and scenario–reason positive rates vary substantially along both dimensions. In contrast, the user-profile–reason surface is dominated by

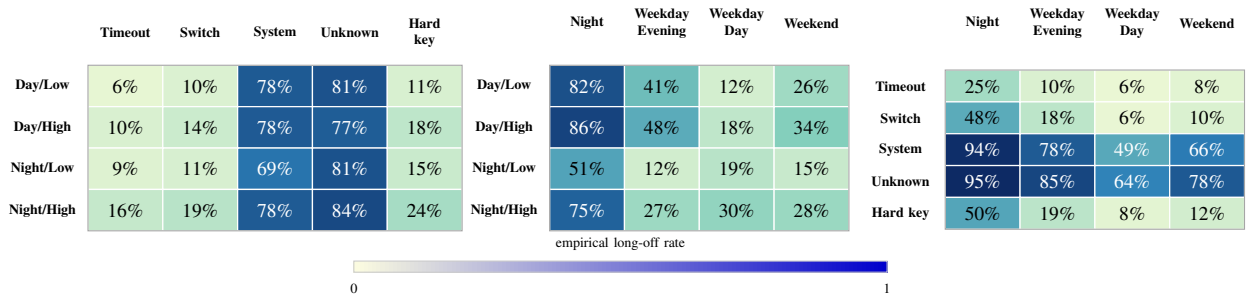


Fig. 1. Positive rates across pairwise interaction contexts. Cell values and colors indicate the proportion of screen-off intervals exceeding four hours. The four user-profile proxies are formed by crossing day/night activity preference with low/high historical long-off rate. Only training and validation events are used.

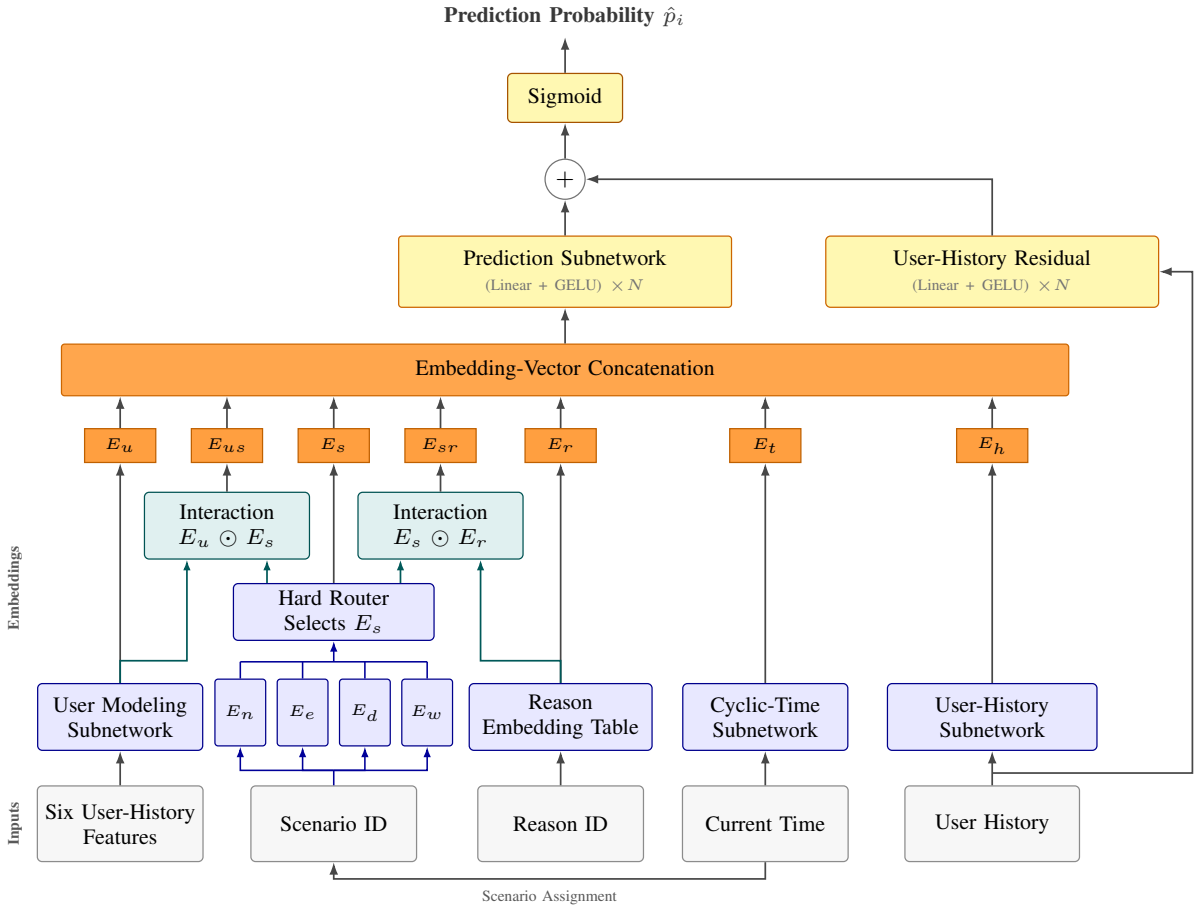


Fig. 2. PRISM architecture. E_u, E_s, E_r, E_t, and E_h denote the user-profile, scenario, reason, cyclic-time, and history embeddings. The interactions are $E_{us} = E_u \odot E_s$ and $E_{sr} = E_s \odot E_r$; E_n, E_e, E_d, and E_w represent the four scenario embeddings. N is the number of Linear–GELU blocks, and $\hat{p}_i$ is the predicted long-off probability.

reason, with only small differences among user-profile groups within each reason column. PRISM therefore retains

$$E_{us} = E_u \odot E_s, \qquad E_{sr} = E_s \odot E_r, \tag{4}$$

while omitting the weaker $E_u \odot E_r$ interaction.

User history embedding. For event i, PRISM constructs three 43-D history vectors. h_u summarizes all completed prior events of user u; h_{u,r_i} uses only that user's prior events with the same screen-off reason r_i as the current event; and h_{u,s_i} uses only prior events occurring in the same scenario s_i. All three vectors contain the same 43 statistics, covering overall behavior, recent activity, and temporal patterns. Unlike the embedding-level Hadamard products introduced above, these vectors consist of statistics computed directly from completed past events.

The three history vectors are concatenated and mapped through a $129 \to 16 \to 16$ subnetwork with LayerNorm, GELU, and dropout to produce the user-history embedding E_h.

Final prediction. PRISM concatenates the five base embeddings and two interaction vectors:

$$z_i = [E_u, E_s, E_r, E_{us}, E_{sr}, E_t, E_h] \in \mathbb{R}^{112}. \tag{5}$$

The prediction head maps z_i to the main prediction logit. In parallel, the user-history residual maps the 129-D concatenated

Table 1: Overall results. We evaluate Bayes, LightGBM, Transformer, and PRISM in predictive performance, model size, and single-event inference time. Immediate and delayed prediction invoke the model at screen-off and after a scenario-dependent delay, respectively.

Model	Acc.	Prec.	Rec.	F1	Macro F1	Model size	Time (ms)
Immediate							
Bayes	86.87	78.62	82.07	80.31	77.36	**44**	**0.012**
LightGBM	88.10	80.32	84.15	82.19	78.99	177,100	0.721
Transformer	88.23	80.97	83.57	82.25	79.13	36,729	1.316
PRISM	**88.71**	**81.72**	**84.24**	**82.96**	**79.87**	35,346	0.394
Delayed							
Bayes	91.04	**88.18**	83.77	85.92	83.72	**44**	**0.012**
LightGBM	92.85	87.41	91.21	89.27	87.97	177,100	0.724
Transformer	92.71	87.74	90.26	88.98	87.63	36,729	1.309
PRISM	**92.89**	86.34	**92.90**	**89.50**	**88.31**	35,346	0.395

history vector to a residual logit, preserving a direct prediction path for information that may be lost during its compression into E_h. The two logits are added before the sigmoid, and the resulting probability is converted to a binary prediction using the validation-selected global threshold θ:

$$\hat{p}_i = \sigma\left(g_{\text{main}}(z_i) + g_{\text{res}}(h_i)\right), \qquad \hat{y}_i = \mathbb{I}[\hat{p}_i \geq \theta]. \tag{6}$$

V. Experiments

Our experiments address four research questions:

- **RQ1:** Does PRISM improve predictive quality while remaining compact and fast?
- **RQ2:** Are PRISM's design choices better than their alternatives?
- **RQ3:** Do PRISM's inputs reflect observed behavioral differences?
- **RQ4:** Where does PRISM improve over the baselines?

A. RQ1: Does PRISM improve predictive quality while remaining compact and fast?

We compare PRISM with three independently tuned baselines. Categorical Bayes uses three categorical inputs: the current screen-off reason, the four-scenario assignment, and a five-category user group derived from completed history. LightGBM receives the same 133 scalar features used by PRISM's user-history and cyclic-time branches. The Recent Transformer encodes up to the 64 most recent completed events of each user, representing each event by its duration, outcome, cyclic timestamp, screen-off reason, and observed next-screen-on reason. It combines the resulting sequence representation with the visible reason and cyclic timestamp features of the current event.

All models use the same chronological training, validation, and test partitions. For each model, validation selects the hyperparameters, number of training epochs where applicable, one global decision threshold, and the training-data selection rule. Candidate rules determine whether training uses all events or only events with at least four completed prior events and, under delayed prediction, whether training is restricted to post-delay candidates. These alternatives are evaluated separately for each model, and the cohort with the best validation performance is retained. The selected model configuration is then refit on the combined training and validation partitions, corresponding to the first 80% of the timeline. The validation-selected threshold is retained, and the test set is used only for final reporting.

Table 1 compares predictive performance, model size, and single-event inference time. Model size denotes stored probability values for Bayes, total tree nodes for LightGBM, and trainable parameters for neural models.

PRISM achieves the highest observed accuracy, F1, and macro-scenario F1 under both protocols, with lower single-event inference time than Transformer and LightGBM.

B. RQ2: Are PRISM's design choices better than their alternatives?

We compare each design choice with its alternatives while keeping all other model components and training settings unchanged.

Table 2: Scenario representations. No scenario omits the scenario representation; Four scenarios selects among learned embeddings for Night, Weekday Day, Weekday Evening, and Weekend; Continuous scenario maps cyclic time features to a continuous scenario vector; and 336 slots learns a separate embedding for each half-hour weekly slot.

Variant	Immediate F1
No scenario	82.8339
Four scenarios	**82.9617**
Continuous scenario	82.9352
336 half-hour slots	82.8057

Scenario. Because delays depend on scenario, alternatives are compared only for immediate prediction. Four scenarios performs best, separating regimes without excessive fragmentation and enabling different delays.

Table 3: User-profile representations. Continuous user-profile maps history features to E_u; Hard four-way crosses day/night activity preference with low/high historical long-off rate.

Variant	Immediate F1	Delayed F1
Continuous user-profile	**82.9617**	**89.5011**
Hard four-way	82.9317	89.4105
One shared bucket	82.9596	89.4952
No user-profile	82.9151	89.4705
Private user id	82.7740	89.3581

User profile. The continuous representation retains graded differences in users' activity, long-off tendency, and recency. It therefore performs better than imposing hard group boundaries, sharing one profile across users, removing the profile, or learning private user-id embeddings.

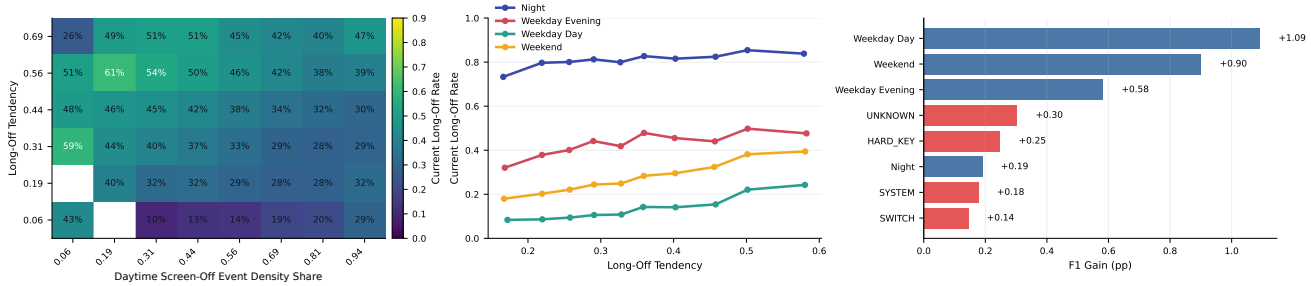


Fig. 3. Positive-rate patterns and PRISM's contextual F1 gains. Daytime screen-off event density share is $(N_d/16)/(N_d/16 + N_n/8)$, where N_d and N_n are a user's prior daytime and nighttime screen-off counts. Long-off tendency is the exponentially weighted historical proportion of intervals exceeding four hours; current long-off rate is the observed proportion in each plotted group. Heatmap values are percentages. F1 gain is PRISM's F1 minus the higher F1 of LightGBM and Transformer within each group, measured in percentage points (pp). Only training and validation events are used in the left and middle panels. Cells containing fewer than 100 events are left blank.

Table 4: Interaction ablation. The symbols E_u, E_s, and E_r denote the user-profile, scenario, and reason embeddings, respectively.

Variant	Immediate F1	Delayed F1
None	<u>82.9603</u>	89.4584
$E_u \odot E_s$	82.9388	89.4882
$E_u \odot E_r$	82.9063	<u>89.4898</u>
$E_s \odot E_r$	82.9312	89.4895
$E_u \odot E_s$, $E_u \odot E_r$	82.9562	89.4866
$E_u \odot E_s$, $E_s \odot E_r$	**82.9617**	**89.5011**
$E_u \odot E_r$, $E_s \odot E_r$	82.9543	89.4406
All three	82.9420	89.4897

Interaction. Among all eight subsets, the selected subset $\{E_u \odot E_s, E_s \odot E_r\}$ achieves the best F1 under both protocols. This agrees with Figure 1: positive rates vary along both axes for user-profile–scenario and scenario–reason, whereas user-profile–reason is largely reason-driven.

Table 5: User-history residual. With residual and No residual include and omit the direct branch from the uncompressed history vector to the prediction logit, respectively.

Variant	Immediate F1	Delayed F1
No residual	82.6961	89.3144
With residual	**82.9617**	**89.5011**

User-history residual. The direct path preserves predictive information from the uncompressed history vector h_i that may be weakened by its compression into the 16-D embedding E_h. The residual improves F1 under both protocols, confirming the benefit of retaining this uncompressed history signal.

C. RQ3: Do PRISM's inputs reflect observed behavioral differences?

Figure 3 analyzes immediate-prediction events using only information available at prediction time. The left panel shows that daytime activity and historical long-off tendency jointly distinguish current long-off rates, supporting history-derived user representations. The middle panel shows that historical long-off tendency is associated with current risk and that this relationship varies across scenarios, supporting both history statistics and scenario-aware modeling. Together with the patterns in Figure 1, these results support PRISM's history, scenario, and reason inputs.

D. RQ4: Where does PRISM improve over the baselines?

The right panel of Figure 3 shows that PRISM outperforms the strongest baseline across all four scenarios. Its largest scenario-level gains occur in Weekday Day and Weekend, whereas the gain for Night is smaller because the strongest baseline already achieves a high F1. At the reason level, PRISM gains more on UNKNOWN and HARD_KEY than on SYSTEM and SWITCH. Overall, PRISM provides the greatest benefit in harder-to-predict contexts, where history, scenario, and reason need to be modeled jointly.

VI. Conclusion and Future Work

We presented PRISM, a personalized compact model for long screen-off prediction using current-event information and completed user history. PRISM achieves the best accuracy and F1 under both immediate and delayed predictions while remaining small and fast. Its largest gains occur in harder-to-predict contexts, highlighting the value of combining representations of user profile, scenario, reason, time, and user history with selected embedding interactions.

Because our evaluation is offline, F1 and inference time do not directly measure on-device energy savings. Future work should evaluate energy and latency on real devices, test generalization across devices and firmware versions, and optimize decision thresholds for deployment costs.

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